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CyDERMS: Center for Cybersecurity & Resiliency of DER & Microgrid-integrated Distribution Systems

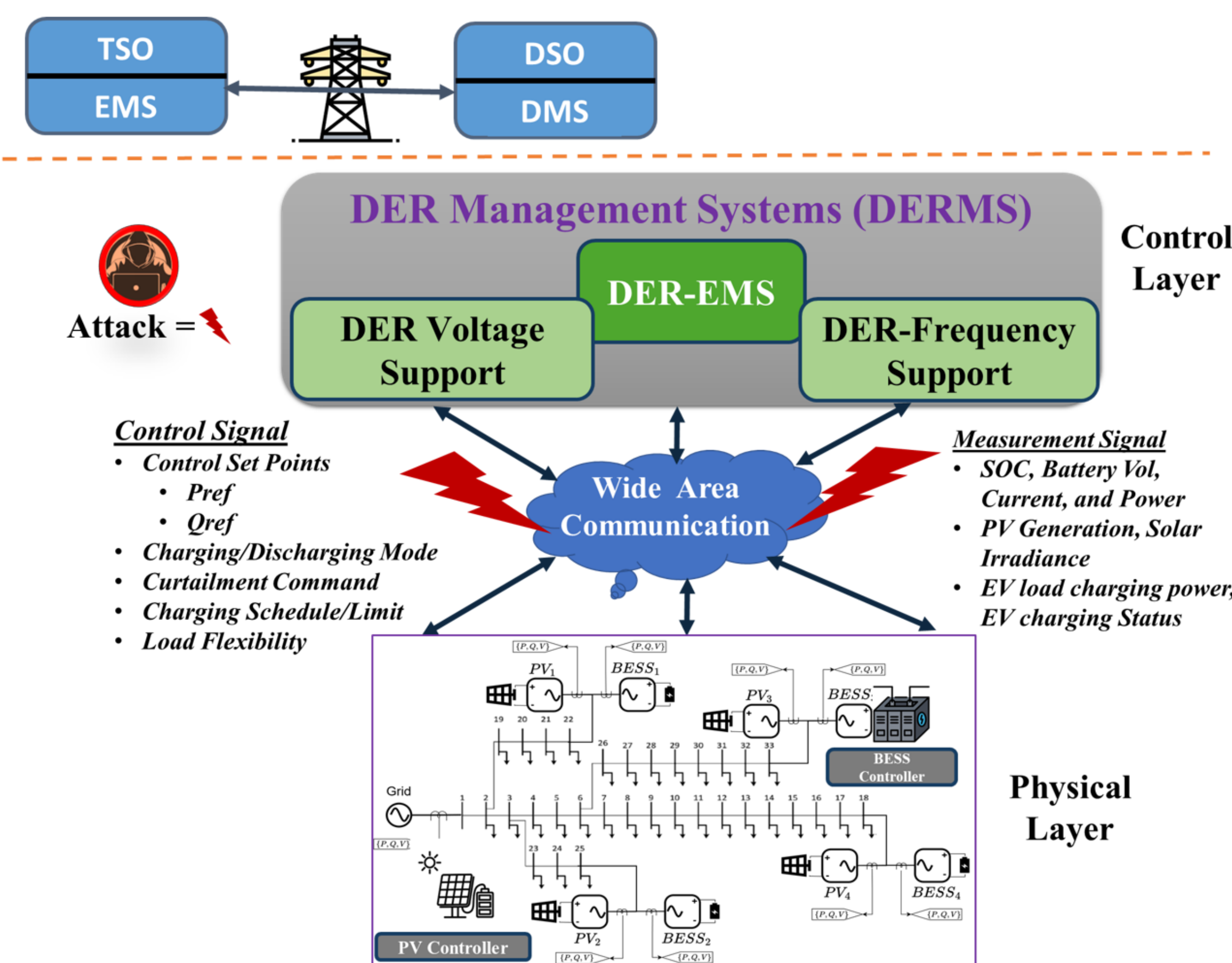
Motivation & Objective

- DER and BESS growth make **DERMS** essential for real-time monitoring, dispatch, and secure control.
- In Iowa, EIA reported nearly **13,800 MW** of utility-scale renewable capacity in early 2025.
- FERC Order 2222** enables DER participants and aggregators to join wholesale electricity markets, increasing the need for secure DER coordination.
- More grid-edge controllers, protocols, and real-time data expand the cyberattack surface.
- Attackers can target control signals such as **Pref**, **Qref**, and charging commands.
- False control or measurement data can affect DER dispatch, voltage/frequency support, and system operation.
- CyDERMS** improves DERMS cyber-resiliency with standards-based integration, including **IEEE 1547** and **IEEE 2030.11**.

Vulnerabilities of DER Communication Protocols

Vulnerability to Attack	SunSpec Modbus	IEEE 1815 (DNP3)	IEEE 2030.5 (SEP2)
Replay	X	X	X
DoS	X	X	X
Eavesdropping	X	X	X
Spoofing	X	X	X
Data Modification	X	X	X
MitM	X	X	X

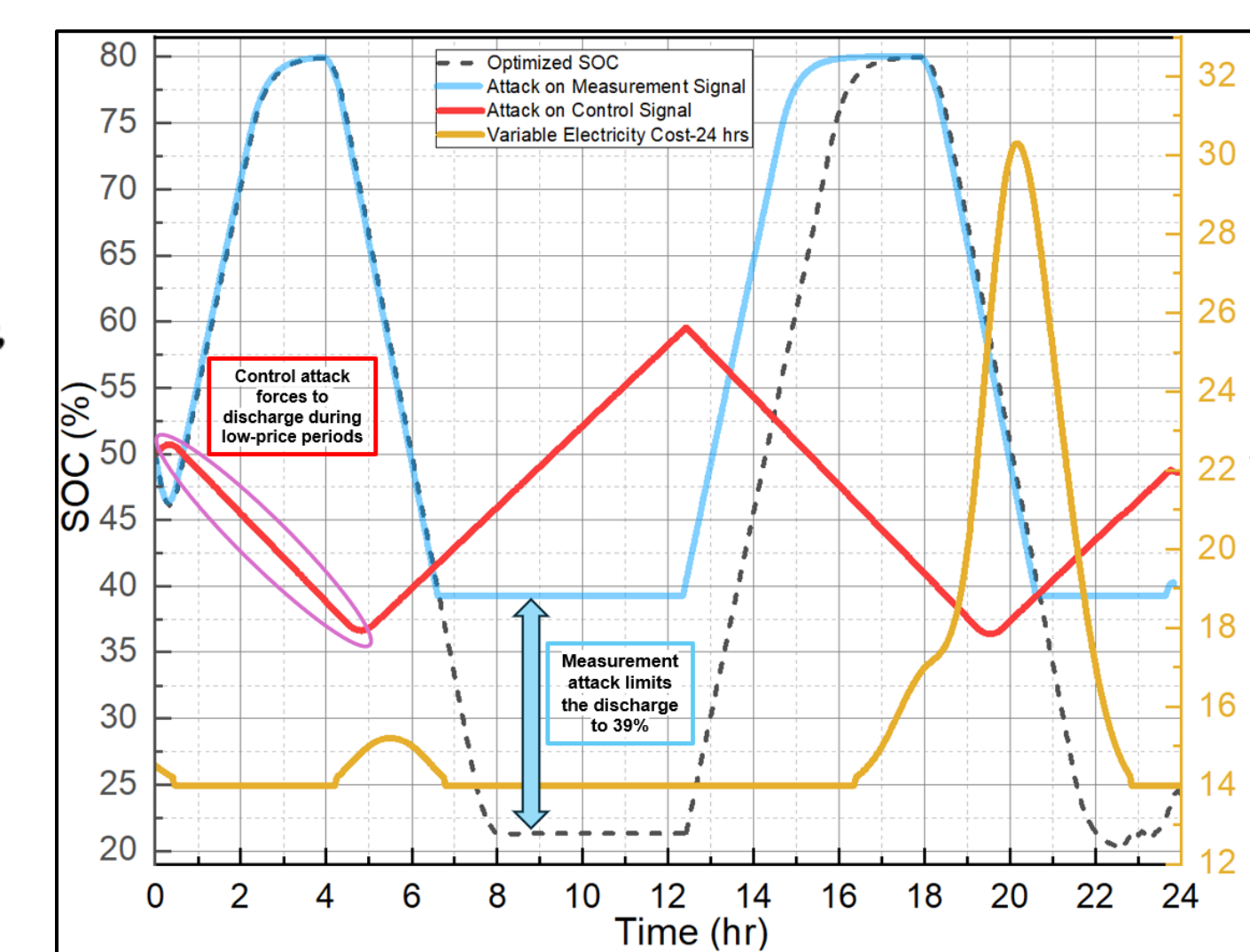
Cyber-Physical DERMS



Cyberattack Impact Analysis

Attack Scenario mapped with the NESCOR framework

Attack Target	Attack Scenarios	NESCOR Mapping	Vulnerabilities	Possible Impact
BESS 1 (Single Attack)	Malware (Scenario 1 & 2)	DER.3 Malware introduced in the DER system during deployment	Insider threats, unauthorized firmware installations, weak credentials	Financial loss to utility and DER owner, large reverse power flow
BESS 1 (Single Attack)	FDIA (Scenario 4)	DER.16 DER management system issues invalid commands	Unauthorized SCADA modifications and harmful commands	Power outages, financial loss, equipment damage, potential loss of life from power surges
BESS 1 & 2 (Coordinated Attack)	FDIA (Scenario 3 & 5)	DER.17 Utility DERMS miscalculates DER energy/service	Unauthorized changes threaten DERMS power flow accuracy and reliability	DER systems profit at utilities' expense, leading to legal disputes and associated costs

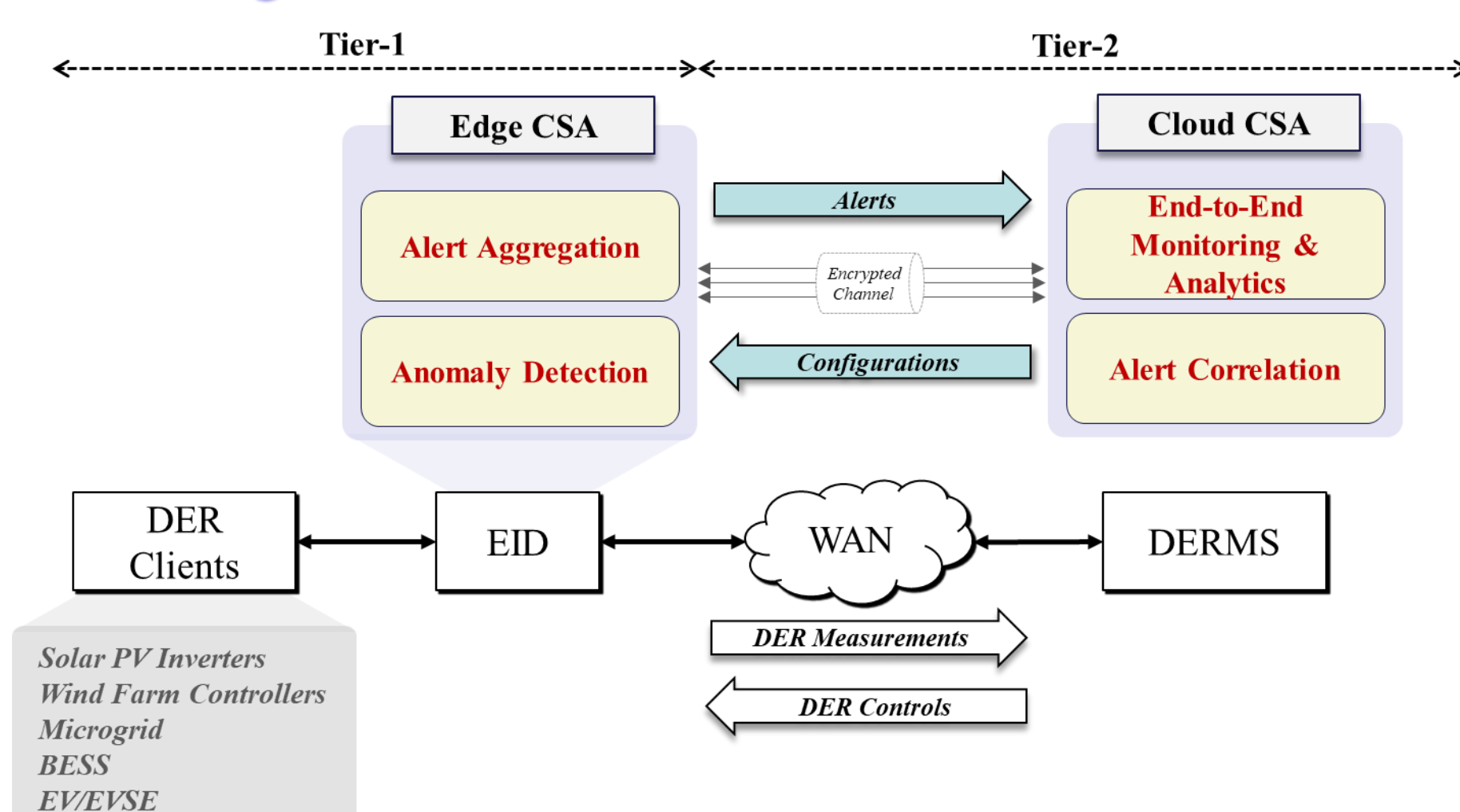


Operational 24hr Summary for Each Scenario

Scenario No.	Cost per day (\$)	System Loss (MWh)	Operational Payout (\$)		
			BESS 1	BESS 2	Total
Base Case	6302.39	1499	21.62	21.62	43.24
1	6363.61	1501	-45.48	33.23	-12.25
2	6352.14	1523	-46.47	33.26	-13.21
3	6367.97	1523	3.47	-13.14	-9.67
4	6359.78	1506	-36.71	32.19	-4.52
5	6372.81	1494	-11.30	-11.30	-22.61

- No benefit of having energy storage at the DS level
- The number of charge-discharge cycles is higher
- Reducing the long-term impact on the investment returns for the BESS owner, along with the running cost-benefit.

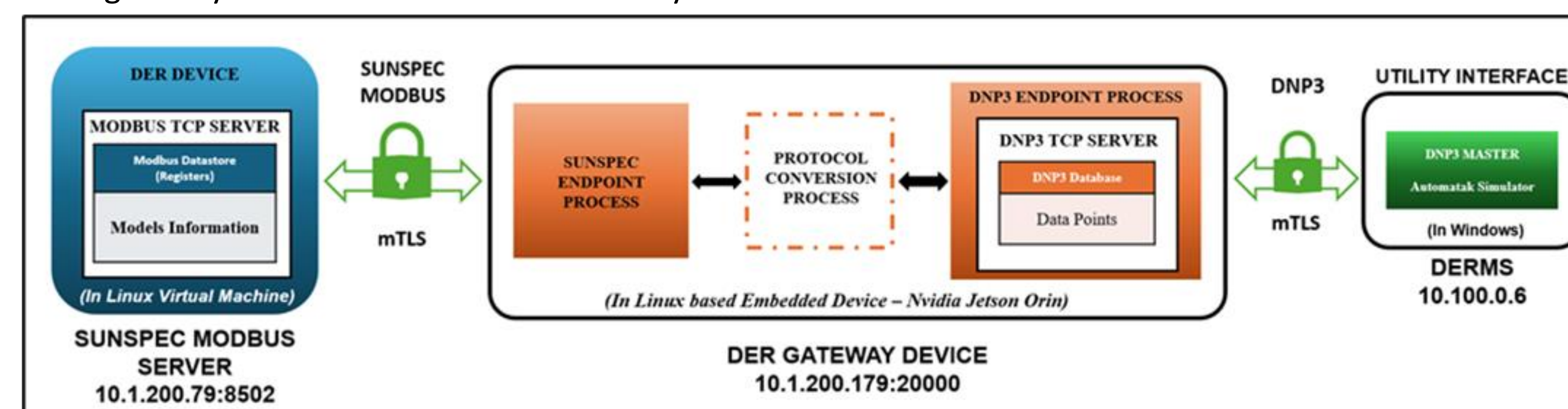
DER Cyber Situational Awareness



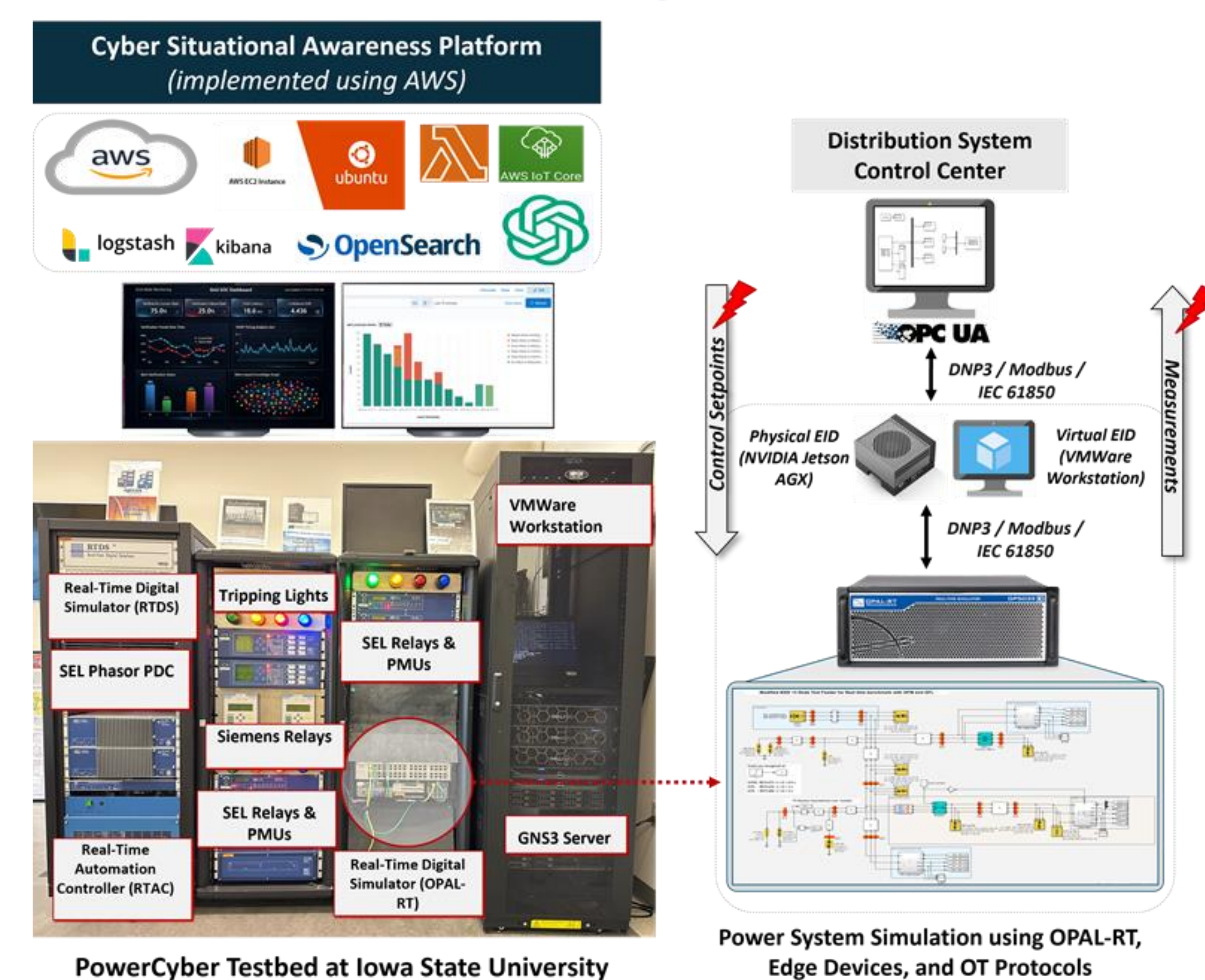
- A two-tier DER architecture is developed for monitoring the DER network for attacks, both in measurement and network features.
- The Edge CSA has both Anomaly Detection and Alert Aggregation algorithms.
- The Alerts are then sent to our Cloud CSA for end-to-end monitoring and alert correlation.

DER Interoperability Testbed

- Developed a functional **SunSpec-compliant Modbus Server** with CIMs including 701, 704, 705, and 711 models.
- Developed a functional DER Network with Protocol Converter between **SunSpec Modbus** and **DNP3** for interoperability and mutual TLS for Cyber Security
- DER gateway and DER Device are secured by TLS 1.2 to mask Modbus and DNP3 Protocol Vulnerabilities.

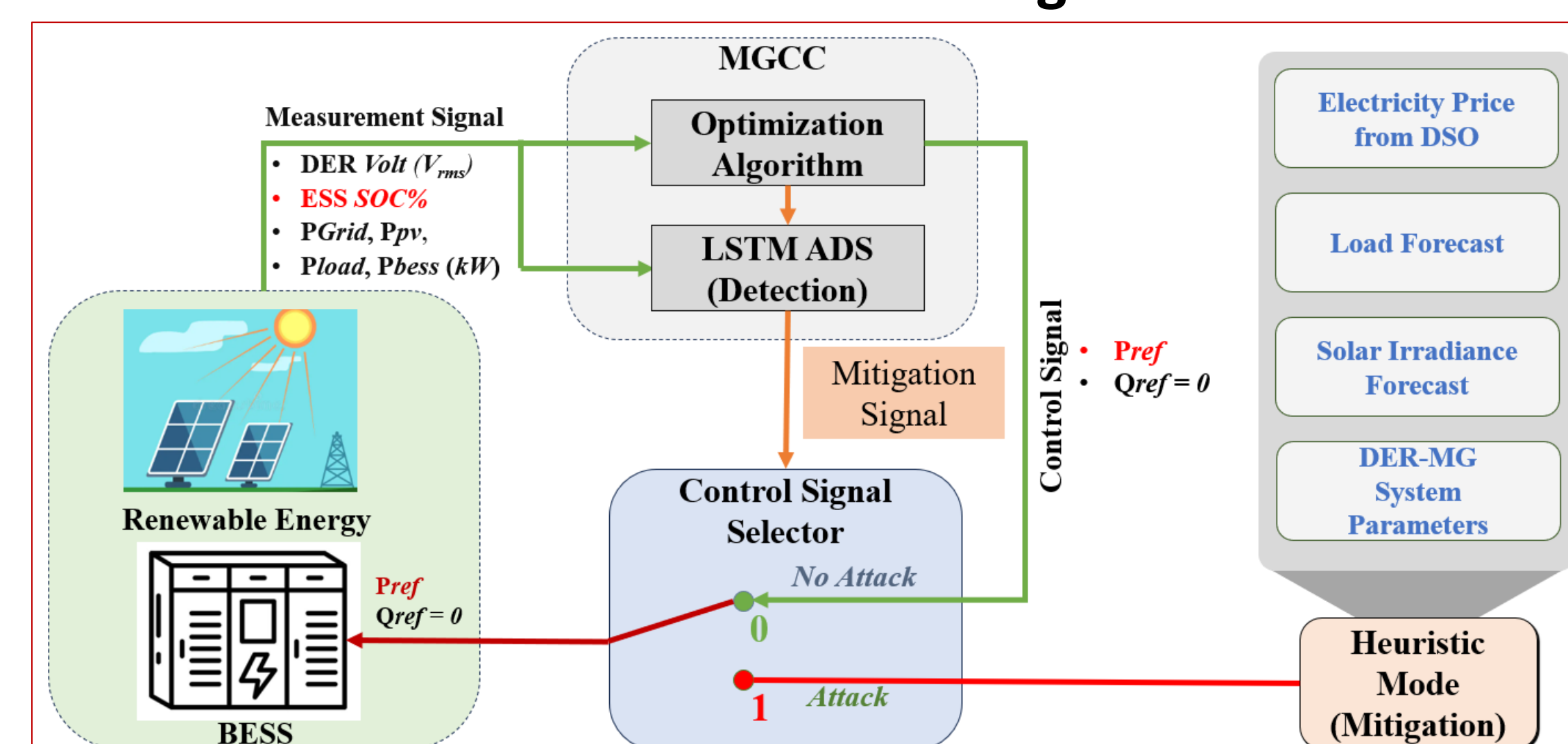


DERMS CPS PowerCyber Testbed at ISU

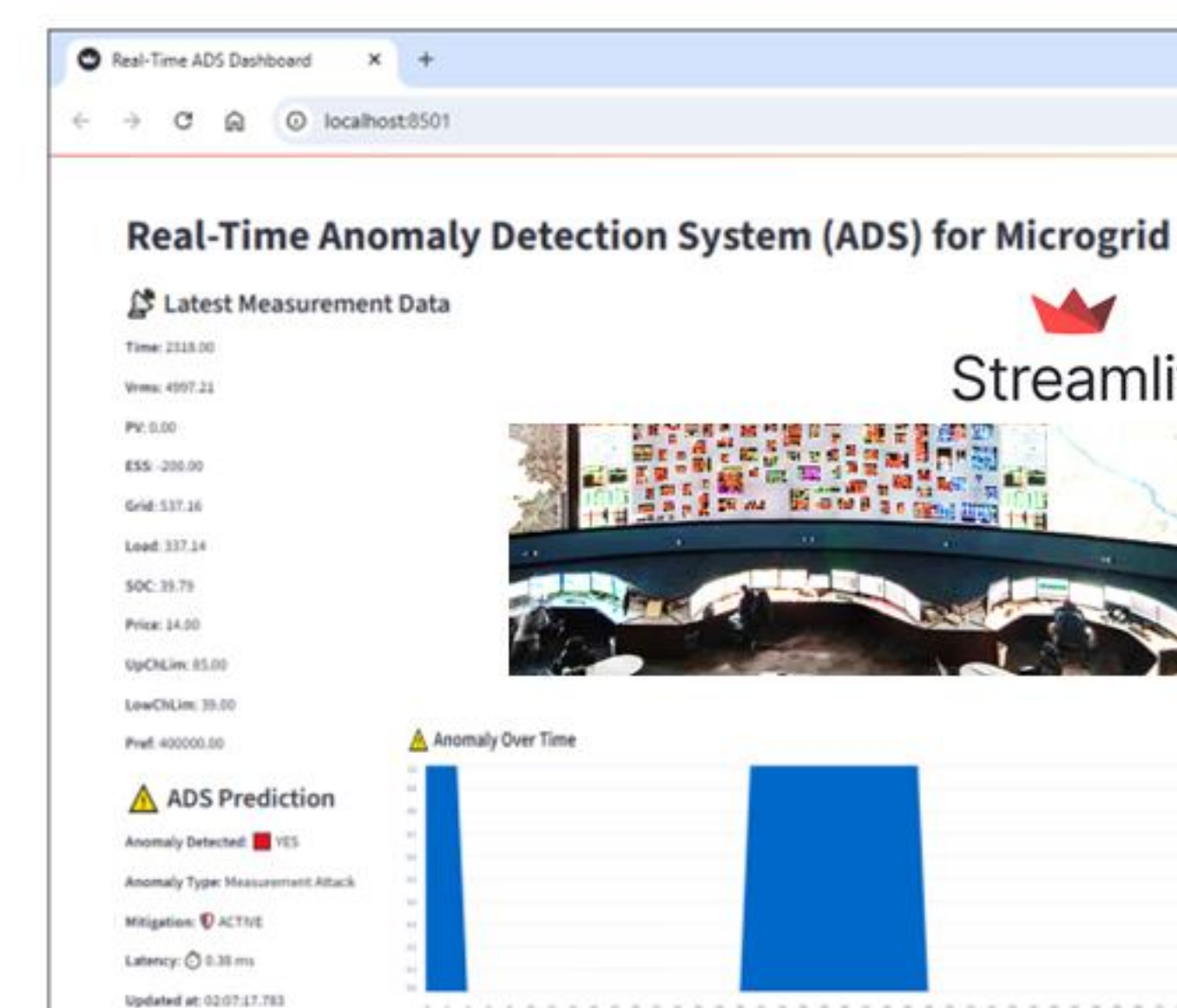
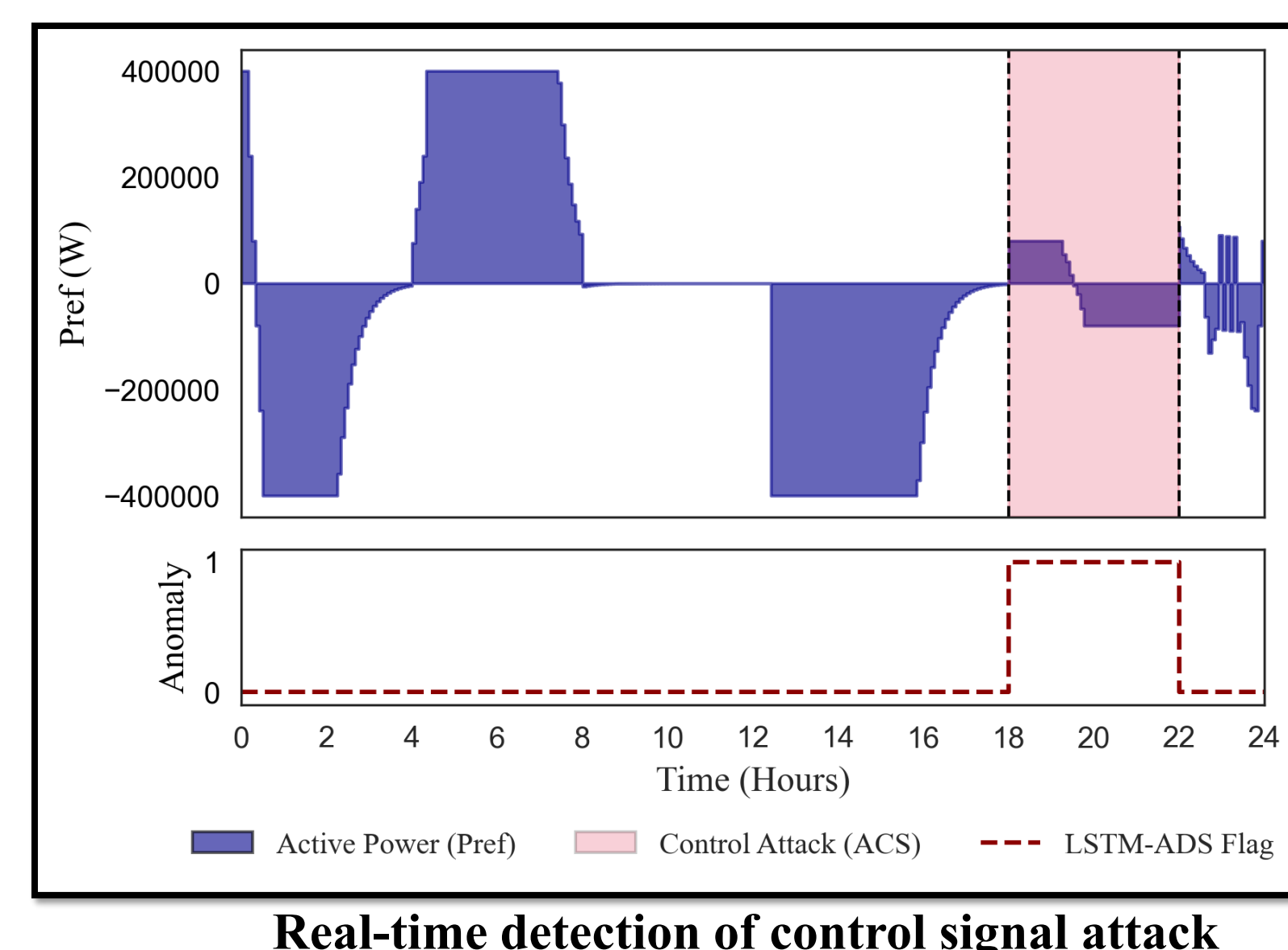


Attack Resilient Control for Microgrid

ARC = Detection + Mitigation



Real Time Validation



Real-time web-based ADS dashboard for microgrid operations implemented in the MGCC workstation running in a Windows VM

Conclusion

- Edge-level anomaly detection and alert aggregation with cloud-level monitoring and alert correlation.
- SunSpec Modbus-to-DNP3 protocol conversion using JSON mapping and MATLAB-based DERMS monitoring/control.
- NESCOR-based DERMS attack modeling and evaluation of operational and cost impacts.
- Online LSTM-based ADS with fallback heuristic mitigation using historical data.
- Contributions evaluated using the ISU-PowerCyber HIL setup for realistic DERMS cyber-physical testing.

